

RailMind: Multimodal Reasoning for Railway Infrastructure Security

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Abstract—Railway systems are operationally critical infrastructures, where minor track anomalies, foreign object debris, and component deterioration can escalate into operational disruptions or system integrity compromises if left undetected. Unmanned aerial vehicle (UAV) inspection provides a practical way to support large-scale railway infrastructure security, yet reliable deployment remains difficult under long-tailed defect distributions, complex outdoor conditions, and the high cost of dense anomaly annotation. Existing surveillance anomaly detection and critical-infrastructure security studies highlight the importance of early risk identification, but railway UAV imagery poses additional challenges because anomalies are often small, irregular, and strongly tied to domain-specific component functions. We propose RailMind, a lightweight railway security inspection framework that pairs domain-specific multimodal reasoning with fine-grained visual grounding. The design centers on a fine-grained domain distillation pipeline: (i) domain-aware reasoning distillation from a closed-source teacher into a deployable student, (ii) fine-grained perception fusion via a language-conditioned detector that grounds textual risk cues as bounding boxes, and (iii) generative data augmentation with optional reinforcement learning on synthetic samples carrying deterministic labels to mitigate data scarcity and tail-class imbalance. Together, these modules facilitate multi-scenario defect identification, spatial localization, and safety-oriented reliability analysis for railway infrastructure. We further introduce a multi-scene, open-vocabulary UAV railway benchmark with aligned protocols for detection and image-level classification, on which prior MLLMs show limited reliability. With only 8B parameters, RailMind achieves state-of-the-art anomaly detection, improving mean precision by $\sim 31\%$ and mean recall by $\sim 19\%$ over the strongest competing model (IoU = 0.5). For image-level defect classification, RailMind-8B improves macro-F1 by $\sim 37\%$ over the best same-scale open-source baseline and is comparable to the strongest closed-source Gemini model. We provide a reproducible pipeline and will release code, data, and models to support future research on trustworthy railway infrastructure security.

Index Terms—Railway infrastructure security, UAV inspection, multimodal large language models, anomaly detection, visual grounding, critical infrastructure security.

I. INTRODUCTION

RAILWAY systems are critical infrastructures that handle large passenger and freight volumes, connect urban and regional transportation networks, and support economic and emergency response. Their security is therefore not limited to cyber defense or access control, but also depends on continuous monitoring of the physical environment: missing fasteners, loosened bolts, corrosion, foreign objects, bird nests, plastic debris, and debris accumulation can all become safety-critical security hazards when they affect tracks, bridges,

catenary systems, transmission towers, or surrounding right-of-way areas. Prior studies on critical-infrastructure security have shown that seemingly component-level anomalies or physical incidents can propagate into larger system-level risks [1]–[4], while railway cyber security surveys further emphasize the need for integrated protection of interconnected railway assets [5], [6]. In this context, visual anomaly detection is not merely a maintenance task, but a first-line monitoring capability for railway infrastructure security.

Unmanned aerial vehicle (UAV) inspection has become an important tool for large-scale railway infrastructure monitoring and protection. Modern railway inspection systems collect high-resolution images along tracks, bridges, electrification structures, and surrounding environments, enabling inspectors to identify defects such as corrosion, missing fasteners, foreign objects, and structural degradation. Compared with fixed cameras or manual patrols, UAVs can cover wide areas, observe elevated or hard-to-reach components, and provide actionable evidence for post-event analysis. However, railway security inspection remains challenging in practice. Field environments exhibit substantial visual variability due to lighting, vegetation, viewpoint changes, and weather conditions, while actual defects are often rare and follow a long-tailed distribution. In addition, obtaining precise annotations for security-related anomalies is expensive and time-consuming, which further limits the scalability of inspection systems.

Most existing railway inspection approaches rely on discriminative object detection models trained on fixed category sets. Although these methods can achieve efficient inference on edge devices, they are typically constrained by closed vocabularies and limited semantic reasoning capability. When encountering unseen anomalies, compound risks, or evolving protection requirements, such systems often fail to provide reliable detection or interpretable explanations. This limitation differs from q w surveillance anomaly detection [7], [8]: railway security requires not only recognizing an unusual visual pattern, but also connecting it to the function of a specific component and the potential operational risk. Recent advances in multimodal large language models (MLLMs) provide a promising alternative. By integrating visual perception with language reasoning, MLLMs can describe complex scenes, infer anomaly categories, and support instruction-following workflows. However, as illustrated in Fig. 1, despite their strong semantic understanding ability, current MLLMs generally lack accurate spatial grounding for small or irregular defects in domain-specific UAV imagery, making them insufficient as standalone security inspection systems.

To bridge this gap, we propose **RailMind**, a lightweight framework for railway infrastructure security that synergizes domain-specific multimodal reasoning with fine-grained visual grounding. RailMind is designed to handle open-vocabulary

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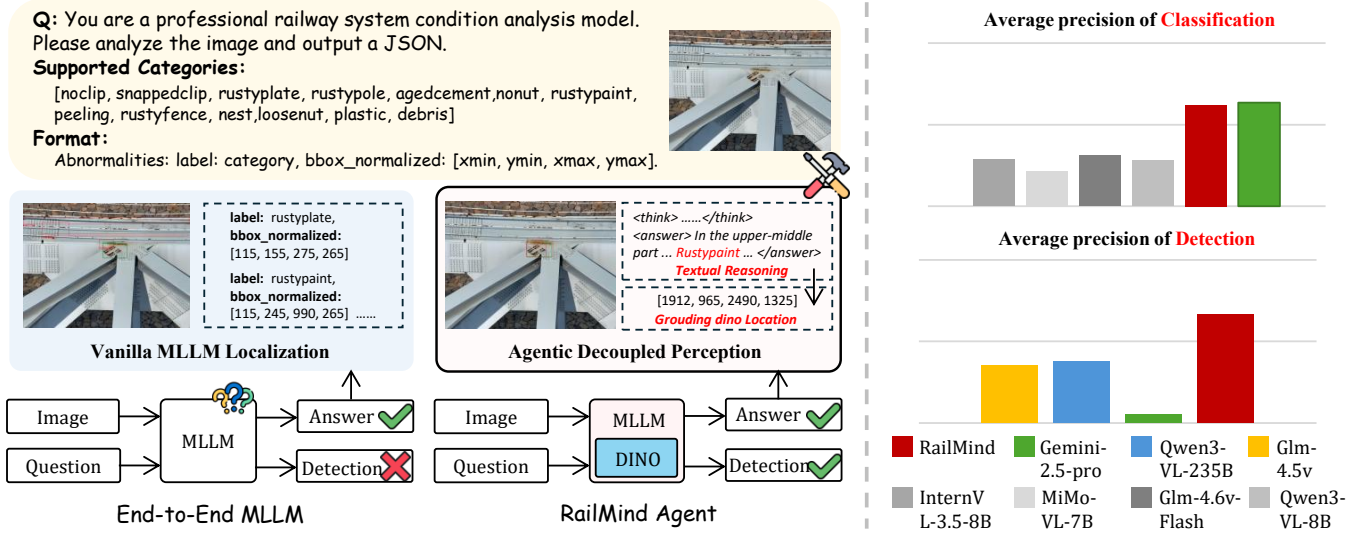


Fig. 1. Overview of the RailMind Agent and benchmark performance. The proposed framework addresses the localization limitations of vanilla end-to-end MLLMs by decoupling anomaly perception into explicit semantic reasoning and specialized visual grounding. As demonstrated on the right, this reasoning-guided approach yields state-of-the-art precision in anomaly detection, while maintaining highly competitive classification performance comparable to powerful closed-source models.

railway anomalies across diverse operational scenarios and to produce evidence that is useful for downstream protection workflows, including anomaly localization, category-level diagnosis, and risk-oriented interpretation. Instead of relying on a monolithic end-to-end MLLM, it decouples semantic reasoning from precise localization, allowing the system to inherit the reasoning capabilities of advanced MLLMs while remaining deployable and efficient.

At the core of RailMind is a fine-grained domain distillation strategy comprising domain-aware reasoning distillation, fine-grained perception fusion, and a generative data augmentation scheme. Specifically, we build a Gemini-driven data engine that injects railway knowledge into a powerful teacher MLLM to generate expert-level supervision, including scene parsing, component-function analysis, anomaly diagnosis, and risk assessment. The resulting supervision is then used to fine-tune a lightweight Qwen3-VL student. For precise localization, RailMind couples the primary MLLM with a Grounding-DINO head that converts language-generated anomaly and risk cues into accurate bounding boxes. To further alleviate annotation scarcity and long-tailed defect distributions, we incorporate Banana-based defect synthesis, and optionally apply GRPO/GSPO reinforcement learning on synthetic data with deterministic labels to improve category consistency.

To facilitate model training and evaluation in this research area, we construct a six-scene, 820-image UAV railway benchmark covering rail tracks, power towers, noise barriers, steel bridges, catenary poles, and surrounding environments. The benchmark reflects realistic long-tailed defect distributions and provides aligned evaluation protocols for both anomaly detection and image-level classification. Unlike generic anomaly benchmarks, it focuses on security-relevant railway assets and anomalies whose significance depends on infrastructure context.

Extensive experiments demonstrate the effectiveness of

RailMind. For anomaly detection, RailMind achieves state-of-the-art performance, improving mean precision and mean recall by approximately $\sim 31\%$ and $\sim 19\%$ over the runner-up at IoU = 0.5. For image-level defect classification, RailMind-8B improves macro-F1 by approximately $\sim 37\%$ over the second-best same-scale open-source model, while remaining comparable to the strongest closed-source Gemini model. These results suggest that combining reasoning-driven semantic analysis with specialized grounding is a practical direction for trustworthy railway infrastructure security.

The main contributions are summarized as follows:

- **RailMind agent.** We present a deployable railway security inspection pipeline that couples a Gemini-driven data engine and supervised fine-tuning of Qwen3-VL with a Grounding-DINO head for language-conditioned localization, augmented by Banana-based defect synthesis and optional GRPO/GSPO reinforcement learning on synthetic data with deterministic labels.
- **Security-oriented benchmark and metrics.** We curate a six-scene, 820-image UAV railway test set covering security-relevant infrastructure scenes and long-tailed anomaly categories, with aligned protocols for anomaly detection (precision/recall under IoU) and image-level macro-F1 classification.
- **Empirical gains.** We report state-of-the-art anomaly detection (mean precision + $\sim 31\%$, mean recall + $\sim 19\%$ vs. the runner-up at IoU = 0.5), a $\sim 37\%$ macro-F1 gain over the second-best same-scale open-source model, and classification performance comparable to the strongest closed-source Gemini.

II. RELATED WORK

A. railway infrastructure security

Railway defect detection has evolved from component-specific deep learning validation toward lightweight onboard models and, more recently, cognition-enhanced inspection. Early studies [9], [10] verified the feasibility of CNNs in specific scenarios. With the emergence of strong detectors such as YOLOv4 [11], subsequent works [12], [13] shifted toward lightweight designs and real-time inference under onboard edge-computing constraints. This emphasis improves deployment efficiency but can weaken deep feature extraction and scene understanding, leading to missed detections in complex scenarios such as vegetation occlusion. Beyond visual detection, railway protection must also be understood as part of broader critical-infrastructure security. Railway cyber-security studies have discussed vulnerabilities introduced by interconnected signaling, control, and communication systems [5], while critical-infrastructure analyses show that local incidents and weak security configurations can create cascading operational risks [1], [4]. These studies motivate visual security inspection as a complementary sensing layer: physical anomalies observed by UAVs provide early evidence for risk assessment before failures or intrusions escalate. Beyond model architecture, the scarcity of high-quality training data is another critical bottleneck. Existing public datasets [14], [15] are primarily collected from urban tram and routine railway surveillance environments, while many other datasets remain proprietary or staged [16], [17]; as a result, they do not fully represent complex real-world operating conditions. Although few-shot [18], [19] and generative [20] strategies have been explored, existing systems, including recent multimodal methods [21], [22], largely remain discriminative models confined to pixel-level perception and lack deeper reasoning about component functions, environmental interactions, and security consequences.

B. Multimodal Large Language Models

Recent advances in Multimodal Large Language Models (MLLMs) have demonstrated remarkable capabilities in multimodal understanding, open-ended generation, and cross-modal instruction following. Early efforts, such as LLaVA [23], [24] and Qwen-VL [25], [26], leverage image encoders and projection modules to align multimodal inputs with the language embedding space. Subsequent research has further extended MLLMs [27]–[31] to additional modalities, including video and audio. As MLLMs continue to evolve, an increasing number of studies attempt to adapt these models to domain-specific applications through post-training strategies such as supervised fine-tuning (SFT). In the railway domain, representative works such as TransGPT [32] tailor general-purpose MLLMs to railway scenarios; however, these approaches are often limited to narrowly defined tasks or label-centric adaptation, leaving broader reasoning and generalization capabilities underexplored. Consequently, there is a growing demand for railway-oriented multimodal models that combine domain knowledge, general visual understanding, and reasoning ability to support complex real-world railway applications.

C. Open-vocabulary Object Detection

Open-Vocabulary Object Detection (OVD) formulates object detection as a region-to-text matching problem, enabling detectors to generalize beyond fixed category sets by leveraging large-scale vision–language pre-training such as CLIP [33]. By constructing a unified visual–semantic space, early works including OVR-CNN [34], RegionCLIP [35], and GLIP [36] progressively align object localization with image–text matching, significantly improving zero-shot recognition. However, most existing OVD methods [37], [38] primarily emphasize semantic generalization, while their grounding accuracy and adaptability in domain-specific open-set scenarios remain limited. Grounding-DINO [39] addresses these challenges by adopting a stronger detector architecture and fine-grained multimodal feature fusion, leading to more robust text-guided localization. Owing to its explicit text-conditioned detection paradigm, Grounding-DINO naturally supports the integration of MLLMs, where textual concepts generated by MLLMs can be directly used as semantic prompts for visual grounding, making it well suited as a detection head for domain-specific tasks with language-defined and evolving category semantics.

III. METHODS

In this section, we describe the proposed RailMind framework for railway infrastructure security. We first present the RailMind agent (Sec. III-A), including the data engine for cognitive security annotation, supervised fine-tuning of the MLLM, detector training with optionally augmented data (Sec. III-A), and fine-grained inference for risk localization (Sec. III-A). We then describe Banana-based defect synthesis for data augmentation (Sec. III-B) and reinforcement learning post-training that exploits deterministic labels from synthetic security-relevant samples (Sec. III-C).

A. RailMind Agent

The proposed RailMind Agent is designed to perform automated security inspection and anomaly evidence extraction from railway infrastructure images captured by unmanned aerial vehicles (UAVs). As summarized in Fig. 2, the agent separates semantic risk understanding from spatial grounding: the MLLM analyzes scene context and potential security-relevant anomalies, while the grounding detector provides box-level evidence for downstream review. We detail the data engine, joint SFT and detector training in Sec. III-A, and fine-grained inference in Sec. III-A.

Data Engine. We devise an automated annotation protocol grounded in Chain-of-Thought (CoT), which leverages the advanced reasoning capabilities of MLLMs as a scalable surrogate for traditional manual annotation. This *data engine* produces structured supervision for both real UAV imagery and, when combined with Sec. III-B, synthesized defect samples. Rather than assigning only category labels, it converts visual observations into security-oriented records that include scene context, component functions, anomaly evidence, and risk descriptions. Fig. 3 sketches how automated reasoning

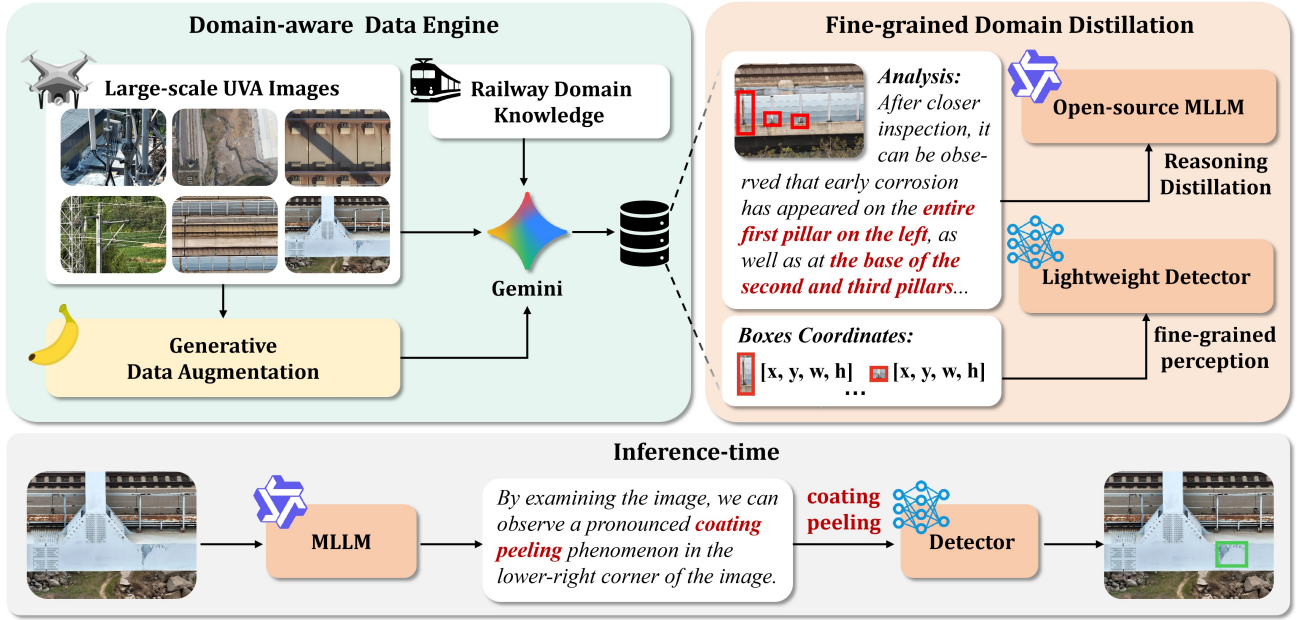


Fig. 2. Overview of the RailMind Agent. In the training stage, UAV images are annotated either automatically via Gemini-2.5-pro or manually, providing SFT data for Qwen3-VL and supervision for Grounding-DINO. During inference, Qwen3-VL first generates textual anomaly descriptions, which are then used by Grounding-DINO to localize anomalies with bounding boxes.

facilitates repository construction to support downstream training.

To ensure high-quality textual supervision for security-oriented annotation, we compare three widely used commercial MLLMs available during dataset construction: GPT-4o [27], Claude-Sonnet [29], and Gemini-2.5-pro [31]. Specifically, representative samples from different railway scenes are paired with identical image-prompt inputs, and each model is evaluated with its advanced reasoning mode enabled. As illustrated in Fig. 4, Gemini-2.5-pro provides more complete and precise anomaly descriptions than the alternatives. While GPT-4o mainly identifies the manually annotated anomaly bounded by the red box and Claude-Sonnet gives only a coarse statement about additional possible defects, Gemini-2.5-pro correctly recognizes both the categories and spatial locations of multiple unannotated yet visually valid anomalies. This observation motivates our use of Gemini-2.5-pro as the teacher model for constructing textual supervision.

Based on this comparison, we instantiate Gemini-2.5-pro as a “Railway Safety Expert Agent” by embedding a knowledge graph—covering railway component functions, defect definitions, and safety implications—directly into the system prompt. This design encourages the model to reason about infrastructure functionality and protection requirements rather than only recognizing visible objects (e.g., “fasteners are designed to secure sleepers”). To mitigate hallucination, we enforce a four-stage reasoning trajectory: (1) *Scene Parsing* decomposes the fundamental component structures within the image; (2) *Function Analysis* verifies the physical state of components against the injected knowledge base; (3) *Anomaly Diagnosis* localizes subtle discrepancies (e.g., trickling rust stains) by contrasting visual features with standard reference states; and (4) *Risk Assessment* deduces potential threats

to operational safety and security based on physical rules. Upon completion, the model yields standardized JSON via a *Format Enforcement Module*, including defect coordinates, fine-grained descriptions, and risk levels, thereby providing rich supervisory signals for downstream multimodal model training.

Supervised Fine-Tuning and Detector Training. During training, UAV images of railway components are processed by Gemini-2.5-pro [31] to generate descriptive annotations of potential anomalies and their security relevance, following the cognitive reasoning pipeline described above. To mitigate data scarcity and long-tailed defect distributions, we further incorporate samples from the Banana-based augmentation in Sec. III-B; those synthesized images likewise pass through the same data engine to obtain image-text pairs before SFT. The outputs from Gemini-2.5-pro are refined and used as supervised fine-tuning (SFT) data for training our main multimodal large language model (MLLM), Qwen3-VL [28]. Formally, the SFT process minimizes the cross-entropy loss between the predicted token distribution and the reference text descriptions:

$$\mathcal{L}_{\text{SFT}} = - \sum_{t=1}^T \mathbf{y}_t \log \hat{\mathbf{y}}_t, \quad (1)$$

where $\mathbf{y}_t$ is the one-hot vector of the ground-truth token at timestep t , and $\hat{\mathbf{y}}_t$ is the predicted probability distribution over the vocabulary.

In parallel, manual annotations specify security-relevant anomaly categories and their corresponding bounding boxes, which are used to train an open-vocabulary detection model based on Grounding-DINO [39]. Given an input image I and a set of textual descriptions $C = \{c_j\}$ representing anomaly or risk categories, Grounding-DINO learns to localize anomalous

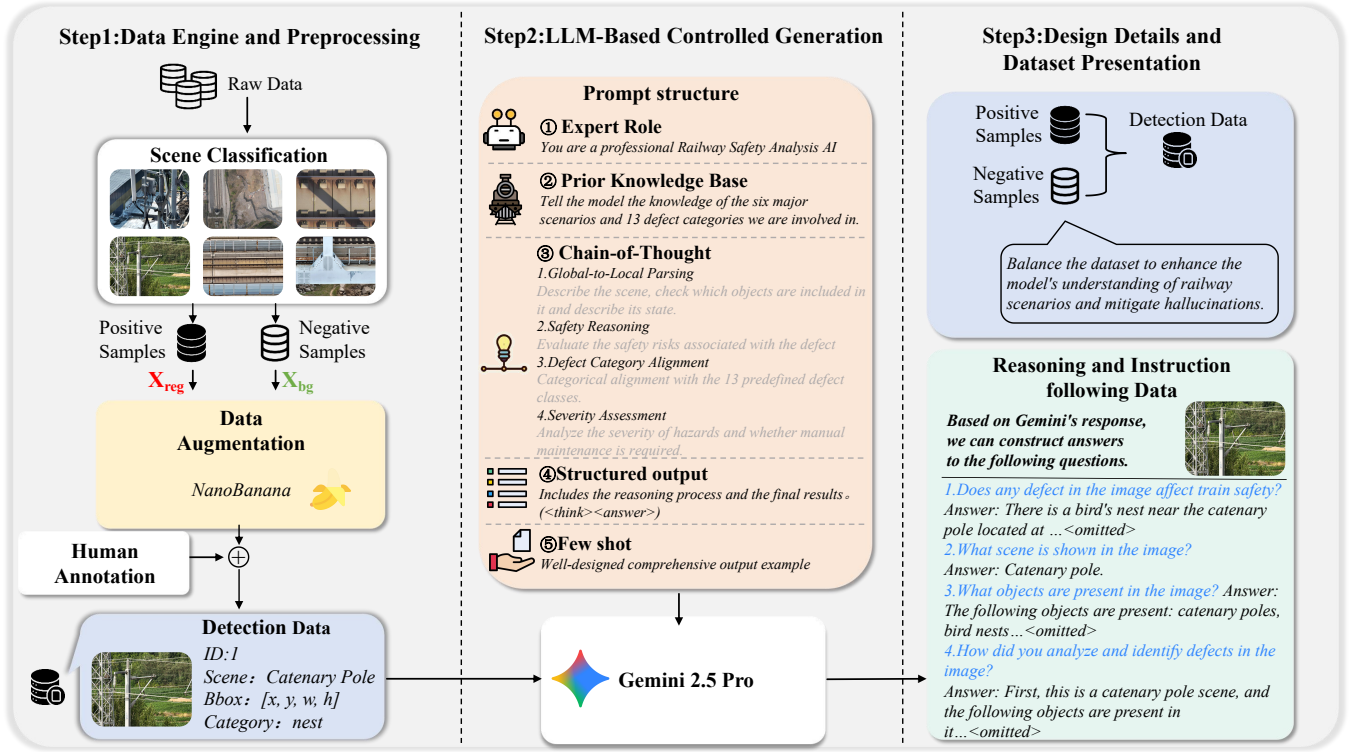


Fig. 3. Overview of the data engine and its role in producing security-oriented semantic supervision for RailMind training, including the annotation pipeline and its integration with augmentation.

regions by predicting a set of bounding boxes $\{\hat{b}_i\}$ and their associated text-conditioned scores.

Following DETR-style detectors [40]–[43], predictions are first matched to ground-truth instances via bipartite matching based on classification and localization costs. The training objective consists of a text-conditioned classification loss and a bounding box regression loss:

$$\mathcal{L}_{DINO} = \lambda_{cls} \mathcal{L}_{cls}^{text} + \lambda_{box} \mathcal{L}_{box}, \quad (2)$$

where $\mathcal{L}_{cls}^{text}$ denotes a contrastive classification loss between predicted object queries and language tokens. Specifically, each object query is projected into the text embedding space and compared with textual features via dot-product similarity, and a focal loss [44] is applied to supervise the matching scores. The localization loss $\mathcal{L}_{box}$ is composed of an ℓ_1 loss and a generalized IoU (GIoU) loss [45] for bounding box regression. Following standard practice in DETR-like architectures, auxiliary losses are applied at intermediate decoder layers and encoder outputs to facilitate optimization.

Fine-Grained Inference. During inference, the trained RailMind MLLM (Qwen3-VL) first analyzes the UAV-acquired image and generates a textual description of observed anomalies, including predicted anomaly categories and risk cues. These textual outputs are then used as prompts for Grounding-DINO, which performs open-vocabulary detection and outputs precise bounding boxes for the detected anomalies. This two-stage pipeline allows RailMind Agent to flexibly produce interpretable security evidence for diverse railway infrastructure defects, even as category descriptions evolve beyond a fixed annotation vocabulary.

B. Generative Data Augmentation

To alleviate data scarcity and long-tailed category distributions in railway security inspection datasets, we develop a targeted data augmentation strategy based on controlled defect synthesis, implemented via **Banana**. This module generates physically plausible and category-consistent anomaly samples under explicit structural constraints, complementing the semantic annotation pipeline of the RailMind agent (Sec. III-A). **Context-Aware Targeted Defect Generation.** Given the highly structured and domain-specific nature of railway security anomalies, generic Text-to-Image (T2I) models [46] often fail to ensure physical plausibility, for example, generating corrosion in semantically invalid regions rather than on metallic components. To address this limitation, we propose an instruction-driven feature transfer strategy that synthesizes defect images under strict semantic, geometric, and textural constraints, as illustrated in the leftmost part of Fig. 3.

Specifically, we first construct a structured feature repository consisting of a background image set X_{bg} and a defect reference set X_{ref} . To avoid semantic inconsistency during synthesis, we enforce category-aligned indexing, requiring X_{bg} and X_{ref} to share the same scene label (e.g., *Steel Bridge*). This constraint reduces cross-domain semantic artifacts and supports category-consistent feature transfer. Building upon this aligned retrieval process, the synthesis procedure is formulated as a conditional generation problem, modeled as $P(I_{syn} | I_{bg}, I_{ref}, T_{inst})$, where T_{inst} denotes structured instruction prompts [47]. These prompts explicitly specify fusion constraints, such as preserving background illumina-

Input Sample	GPT-4o	Claude-opus-4	Gemini 2.5 Pro
Input Image:  Input Prompt: You are a railway safety AI analyzing a railway image captured by a drone. Suspected anomaly areas have been marked with red bounding boxes. If multiple boxes exist, describe each separately. Please complete the following tasks: 1. Identify the type of anomaly, e.g., rust, 2. Describe the physical characteristics of the anomaly, 3. Assess the urgency level of the anomaly and provide a rationale, 4. In addition to the areas within the boxes, are there any other anomalous regions present in the image?	1. Description of Abnormalities Anomaly 1: Corrosion of the guardrail support pillar. Significant reddish-brown rust marks, particularly in the area near the base. Anomaly 2: ... 2. Urgency Level Anomaly 1: Moderate. Due to significant corrosion, the pillar may lose partial structural integrity... Anomaly 2: ... 3. Global Inspection Except for the areas within the bounding boxes, no significant abnormalities were observed in other regions of the image, such as no foreign objects or structural damage on the tracks. ✗	1. Description of Abnormalities Anomaly 1: Severe corrosion of the guardrail pillar. The surface of the pillar exhibits large areas of reddish-brown rust marks, with obvious oxidation patches and uneven coloring. Anomaly 2: ... 2. Urgency Level Anomaly 1: Moderate. Large-scale corrosion may affect the structural strength of the guardrail...Anomaly 2: ... 3. Global Inspection Inspection results for areas other than the labeled boxes: Track Status: Visually good, no obvious cracks or deformation... Other Guardrail Pillars: Signs of minor corrosion; recommended to be included in the routine maintenance plan. ✗	1. Description of Abnormalities Anomaly 1: Severe corrosion of the guardrail/noise barrier pillar. This white metal pillar exhibits large-scale, long-strip severe corrosion.... Anomaly 2: ... 2. Urgency Level Anomaly 1: High. Due to the huge area and severe degree of corrosion, breakage may occur...Anomaly 2: ... 3. Global Inspection Other Guardrail Pillars: Careful observation reveals that another pillar located between Box 1 and Box 2 (the second pillar in the frame), as well as pillars further to the right, also show signs of early-stage corrosion at their bases similar to Box 2. ✓

Fig. 4. Comparison of textual data generation quality among commercial MLLMs. Gemini-2.5-pro provides more complete anomaly discovery and more precise defect descriptions, making it better suited as the teacher model in our data engine.

tion, respecting component geometry, and adhering to surface material properties. Leveraging a multimodal attention mechanism, the generation model extracts high-frequency defect features (e.g., corrosion textures) from the reference image I_{ref} [48], [49] and maps them onto designated regions of the background image I_{bg} while maintaining perspective and geometric consistency. Furthermore, to mitigate the long-tail distribution commonly observed in real-world railway security inspection data, a quantitative task planner dynamically adjusts the synthesis frequency according to the real data distribution D_{real} , assigning higher sampling weights to underrepresented anomaly categories following $W_c \propto 1/N_c$ [50]. This strategy enables targeted generation of extremely scarce risk patterns, such as *Fastener Fractures*, thereby rebalancing the training data distribution.

C. Reinforcement Learning

A key advantage of the synthesized defect data from Sec. III-B is the availability of explicit, deterministic ground-truth (GT) anomaly categories. Because the generative process specifies the defect classes by design, these data avoid much of the ambiguity and noise typically associated with manual human annotation. Consequently, the precise alignment between visual inputs and their corresponding GT labels makes the synthesized dataset naturally suited for Reinforcement Learning (RL) [51] post-training after SFT in Sec. III-A, especially when optimizing category consistency for security-sensitive screening.

Specifically, the synthesized data are first processed by the Gemini-2.5-Pro model to generate descriptive anomaly annotations and risk-aware semantic descriptions, following the same data-engine pipeline as in Sec. III-A. The resulting image-text pairs are used for Supervised Fine-Tuning together with real data, enabling the model to align visual railway defect patterns with structured security descriptions before RL.

Beyond supervised learning, the availability of absolute category-level GT labels allows us to construct a robust reward environment to further enhance classification consistency. We define a set-based reward function $R(C_{pred}, C_{gt})$ that evaluates the agreement between the predicted security-relevant anomaly category set C_{pred} and the ground-truth set C_{gt} :

$$R(C_{pred}, C_{gt}) = \begin{cases} 1.0, & \text{if } C_{pred} = C_{gt}, \\ 0.5, & \text{if } C_{gt} \subset C_{pred}, \\ 0.0, & \text{otherwise.} \end{cases} \quad (3)$$

An exact match receives the maximum reward to encourage precise defect recognition. A partial reward is assigned when the predicted category set forms a superset of the GT set, reflecting a conservative over-prediction preference that is suitable for safety-critical and security-sensitive railway inspection scenarios where missed detections are severely penalized. All other cases receive zero reward.

To systematically optimize the model under this reward structure, we integrate both Group Relative Policy Optimization (GRPO) [52] and Group Standard Policy Optimization (GSPO) [53]. The fundamental distinction between the two lies in how they estimate the advantage function within a sampled

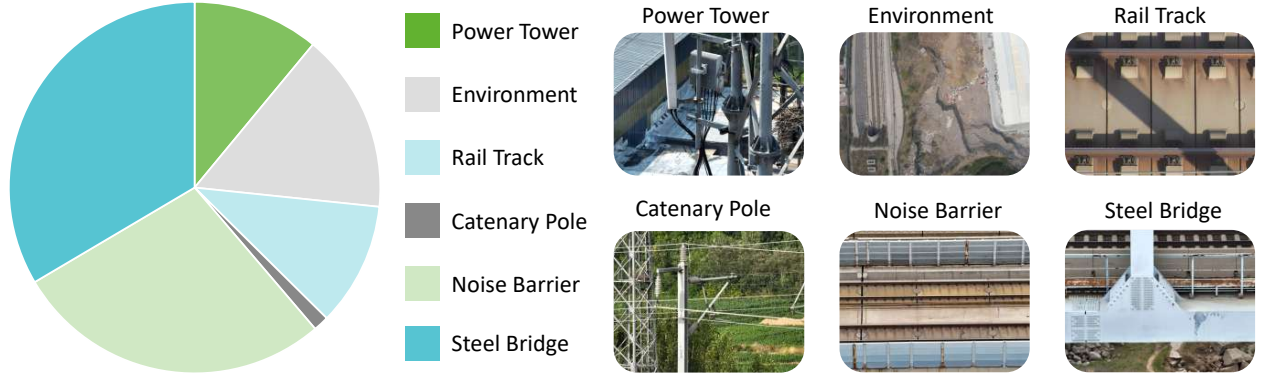


Fig. 5. Test-data distribution across the six security-relevant railway scene categories of the proposed benchmark.

group of G responses for a given prompt. GRPO evaluates the relative advantage by normalizing the rewards within the sampled group:

$$A_i^{\text{GRPO}} = \frac{R_i - \mu_R}{\sigma_R} \quad (4)$$

where R_i is the reward of the i -th response, and μ_R and σ_R represent the mean and standard deviation of the group's rewards. This relative baseline encourages the model to favor the best outputs among its current generative distribution.

Conversely, GSPO evaluates the policy against an absolute, standard baseline V_{ref} (such as a shared value network or a predefined standard score), computing the advantage as:

$$A_i^{\text{GSPO}} = R_i - V_{\text{ref}} \quad (5)$$

By coupling both algorithms, the framework effectively leverages the intra-group relative exploration of GRPO alongside the absolute convergence stability of GSPO. This dual-optimization strategy fully exploits the structured synthetic GT data, significantly improving the semantic accuracy and generalization capability of the model in complex railway environments.

IV. BENCHMARK DATASET

This section describes the multi-scene railway UAV benchmark used to evaluate security-oriented anomaly understanding and localization. We first introduce the construction of the test set and its security-relevant scene categories, and then define the metrics used in the following experiments.

A. Data Construction

To evaluate the performance of our method under realistic railway infrastructure security scenarios, we construct a domain-specific test dataset covering multiple infrastructure scenes and anomaly types. The dataset contains a total of 820 images distributed across six scene categories, including power towers (91 images), catenary poles (11 images), rail tracks (88 images), surrounding environment (128 images), noise barriers (227 images), and steel bridges (275 images). As illustrated in Fig. 5, the dataset follows a long-tailed distribution, reflecting

real-world inspection conditions where certain structures and security-relevant anomalies occur less frequently.

Each scene category is annotated with anomaly types that may affect railway operation, asset integrity, or perimeter security. Rail track images include defects such as missing fasteners and cracked fasteners, which can weaken track stability. Noise barrier scenes cover corrosion of barrier panels and supporting columns, as well as aging and degradation of the mortar layer when visible at a sufficient scale. Steel bridge images contain anomalies including missing bolts, corrosion of steel coatings, coating peeling, and corrosion of bridge guardrails, all of which provide visual evidence of structural degradation. Catenary pole and power tower scenes include foreign object anomalies, primarily bird nests, while power tower images further involve loosened antenna bolts. Environmental scenes focus on surrounding safety hazards, such as plastic film and accumulated garbage, which may threaten railway operation when they intrude into critical areas. For steel bridge and environment scenes, a sparse annotation strategy is adopted, with typically 1–3 bounding boxes per image and rarely more than 5, to emphasize salient anomaly evidence and reduce annotation redundancy.

B. Evaluation Metrics

We assess models along two complementary tracks that reflect different needs in railway security: *anomaly detection* with bounding boxes for spatial evidence localization, and *image-level anomaly classification* based on predicted versus ground-truth defect category sets for rapid risk screening.

Anomaly detection. A predicted bounding box is a true positive (TP) if it matches a ground-truth box of the same anomaly category with intersection-over-union (IoU) above a threshold τ ; otherwise it is a false positive (FP). Unmatched ground-truth boxes are false negatives (FN). This protocol evaluates whether a model can provide localized visual evidence for security-relevant anomalies rather than only textual descriptions. Precision and recall are

$$\text{PR} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{RC} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \quad (6)$$

TABLE I
PRECISION/RECALL OF THE MODEL FOR SECURITY-RELEVANT ANOMALY LOCALIZATION IN RAILWAY SCENARIOS. (IoU=0.5)

Models	Steel Bridge	Rail track	Environment	Catenary Pole	Noise Barrier	Power Tower	Mean
InternVL3-38B	3.6 / 2.4	0.0 / 0.0	1.5 / 0.5	0.0 / 0.0	0.0 / 0.0	0.0 / 0.0	0.9 / 0.5
InternVL3-78B	2.4 / 1.9	1.7 / 0.7	1.7 / 0.7	0.0 / 0.0	0.4 / 0.2	0.0 / 0.0	1.0 / 0.6
Qwen3-vl-8b-instruct	17.7 / 11.4	6.6 / 6.4	9.2 / 2.9	72.7 / 57.1	5.4 / 4.9	9.9 / 8.2	20.3 / 15.2
Qwen3-vl-32b-instruct	27.6 / 20.7	28.2 / 14.3	21.2 / 6.6	85.7 / 85.7	12.6 / 17.5	27.6 / 17.2	33.8 / 27.0
Qwen3-vl-30b-a3b-instruct	15.9 / 10.9	0.0 / 0.0	23.1 / 7.4	76.9 / 71.4	13.8 / 14.5	26.5 / 25.4	26.0 / 21.6
Qwen3-vl-235b-a22b-instruct	24.3 / 16.8	24.1 / 9.3	28.3 / 3.5	91.7 / 78.6	19.1 / 14.5	38.2 / 17.2	37.6 / 23.3
Glm-4.5v	31.5 / 16.2	7.8 / 4.3	21.4 / 2.4	90.9 / 71.4	20.2 / 5.5	40.9 / 14.8	35.5 / 19.1
GPT-4o	0.3 / 0.2	0.0 / 0.0	0.0 / 0.0	8.3 / 7.1	0.0 / 0.0	0.0 / 0.0	1.4 / 1.2
Claude-sonnet-4.5	0.4 / 0.4	0.0 / 0.0	0.5 / 0.3	0.0 / 0.0	0.1 / 0.1	0.0 / 0.0	0.2 / 0.1
Gemini-2.5-pro	3.9 / 7.0	6.2 / 4.3	2.4 / 1.6	13.3 / 14.3	2.8 / 4.5	3.1 / 3.3	5.3 / 5.8
RailMind-Agent-SFT	29.7 / 23.5	58.2 / 50.7	32.0 / 16.5	100.0 / 69.2	66.4 / 57.4	84.9 / 36.9	61.9 / 42.4
RailMind-Agent-GRPO	35.4 / 18.0	83.5 / 72.1	27.5 / 11.7	100.0 / 69.2	68.6 / 55.2	79.2 / 34.7	65.7 / 43.5
RailMind-Agent-GSPPO	39.5 / 17.7	86.2 / 75.7	25.2 / 10.1	100.0 / 61.5	66.7 / 51.0	81.8 / 36.9	66.6 / 42.2

We report per-scene PR/RC as well as the mean across scenes (e.g., Table I). Unless otherwise noted, our main comparison uses the stricter $\tau = 0.5$ setting to evaluate localization accuracy, which is important when detection boxes are used as evidence for inspection review. Table V further ablates $\tau \in \{0.1, \dots, 0.5\}$ under a fixed RailMind–Grounding-DINO setup. For some analyses we summarize detection quality with macro-averaged PR_M , RC_M , and $F1_M$ over categories (Tables IV and V), using the harmonic mean $F1 = 2PRRC/(PR + RC)$ at the same pooling level.

Image-level anomaly classification. For each test image we derive a predicted anomaly category set from the model-generated description and compare it with the ground-truth category set, obtaining per-category precision and recall PR_c and RC_c ; these metrics do not depend on bounding-box matching. This track evaluates whether a model can identify the security-relevant anomaly types present in an image even before precise localization is considered. The F1 score is

$$F1_c = \frac{2PR_c RC_c}{PR_c + RC_c}, \quad (7)$$

and we report macro averages $PR_M = \frac{1}{C} \sum_{c=1}^C PR_c$, $RC_M = \frac{1}{C} \sum_{c=1}^C RC_c$, and $F1_M = \frac{1}{C} \sum_{c=1}^C F1_c$ over C categories (Tables II and III).

V. EXPERIMENTS

A. Experimental Setup

We train the two RailMind components on different hardware platforms. Specifically, the primary multimodal agent is trained on $4 \times$ NVIDIA A800 GPUs, whereas the Grounding-DINO detector is trained on $4 \times$ NVIDIA RTX4090 GPUs. Since the two components serve different roles in the security inspection pipeline—semantic risk analysis and visual evidence localization—we present their data sources and optimization settings separately below.

Primary MLLM agent (Qwen3-VL-8B). The agent is initialized from Qwen3-VL-8B-Instruct and adapted with supervised

fine-tuning (SFT) followed by reinforcement learning (RL). The training data are organized as instruction-response pairs that describe railway scenes, anomaly categories, and risk-relevant observations. For SFT, we combine three instruction datasets containing 2,264, 5,497, and 13,170 samples, respectively, yielding 20,931 training instances in total. We train SFT for 3 epochs with batch size 32 and learning rate 5.0×10^{-6} . For RL post-training, we use 3,134 prompts with batch size 64; for each prompt we sample 16 rollouts to estimate the training signal. RL runs for 1 epoch with learning rate 1.0×10^{-6} .

Grounding-DINO detector. The object detector uses pre-trained weights based on the Swin-T backbone `groundingdino.swint_ogc`, while the text encoder adopts the pre-trained `bert-base-uncased` model. The BERT parameters are frozen throughout training. The training dataset for this component consists of 3,334 images with 10,613 instance annotations, providing box-level supervision for anomaly evidence localization. When constructing text prompts, we use an all-class configuration by default to support open-vocabulary security inspection. The detector is trained for 30 epochs with a batch size of 4. The initial learning rate for the model’s main parameter group is set to 1.0×10^{-4} , whereas the initial learning rates for the visual backbone and linear projection-related parameter groups are set to 1.0×10^{-5} . Optimization uses a step decay strategy (StepLR), reducing the learning rate of each parameter group by a factor of 0.1 every 10 epochs.

B. Results and Analysis

Anomaly detection (bounding-box localization). Table I compares open- and closed-source MLLMs against RailMind on the six-scene benchmark (Sec. IV-A) under the same detection metrics. This evaluation focuses on whether a model can provide localized visual evidence for security-relevant railway anomalies. Among open-source checkpoints, capacity and generalization vary widely: smaller or general-purpose models such as InternVL3-38B/78B remain near zero in

mean precision/recall ($\sim 1\%$), indicating that generic vision-language pre-training alone does not yield usable box-level localization on this UAV railway security task. Stronger open-source models narrow the gap—for example, Qwen3-VL-8B attains 20.3%/15.2% mean PR/RC, Qwen3-VL-32B reaches 33.8%/27.0%, Qwen3-VL-235B reaches 37.6%/23.3%, and GLM-4.5V achieves 35.5%/19.1%—yet their mean precision still trails **RailMind-Agent-GSPO** (66.6% / 42.2%) by roughly 29–35%, and rail-track, steel-bridge, and environment columns remain challenging for many rows (e.g., frequent single-digit or low double-digit PR/RC under cluttered backgrounds or rare defects). The spread across scenes confirms that long-tailed infrastructure, small objects, and complex backgrounds stress both recall and localization consistency in security inspection.

Despite their reputation for general multimodal reasoning, GPT-4o and Claude-sonnet-4.5 exhibit extremely low mean detection scores (1.4%/1.2% and 0.2%/0.1%, respectively), with many scene-wise entries at or near 0%. Gemini-2.5-pro is comparatively stronger among the three APIs but still reaches only 5.3%/5.8% mean PR/RC, far from the RailMind range. These gaps are not explained by a wholesale lack of semantic understanding; the image-level classification results in Table II (Sec. IV-B) make this point quantitative. The contrast between Table I and Table II highlights that *understanding* $\neq$ *grounding*: models with strong category-level reasoning can still fail at box-level localization in domain UAV imagery. For security, this distinction is crucial because inspection decisions require not only an anomaly label but also verifiable spatial evidence.

RailMind-Agent variants achieve the highest mean PR/RC in Table I, with GSPO peaking at **66.6%** / 42.2% and GRPO at 65.7% / **43.5%** recall, substantially exceeding both the strongest open-source MLLMs and all closed APIs. Gains are strongest on rail track, noise barrier, and power tower, where baseline PR/RC often remain in the low tens of percent or below; steel bridge and environment remain relatively harder for all methods, reflecting heterogeneous backgrounds and sparse annotations, though RailMind still improves both metrics over the best competing rows in those columns. These improvements indicate that reasoning-guided grounding is particularly useful when subtle physical evidence must be localized in cluttered security inspection imagery. The catenary-pole column occasionally shows saturated precision for several methods and should be interpreted cautiously given the extremely small test size for that scene (cf. Sec. IV-A). Further image-level classification results (Table II), ablations, and qualitative cases follow below.

Image-level anomaly classification. Building on the detection picture in Table I, Table II evaluates *image-level* security-relevant anomaly category reasoning under the category-set protocol and macro averages in Sec. IV-B (Eq. (7)), without any bounding-box matching. The upper block reports large closed-source APIs and high-capacity open-source MLLMs, while the lower block focuses on comparable-scale open-source models ($\sim 7\text{B}$ – 9B), including RailMind-8B. Large closed-source APIs and high-capacity open-source checkpoints occupy the top block: Gemini-2.5-pro and Claude-sonnet-4.5 reach $F1_M$ of 62.2% and 52.4%, and Qwen3-VL-235B and

TABLE II
MACRO PR/RC/F1 FOR IMAGE-LEVEL ANOMALY CLASSIFICATION.

Models	PR_M	RC_M	$F1_M$
Qwen3-vl-235b-a22b-instruct	63.2	41.5	46.7
Glm-4.5v	<u>67.5</u>	42.9	45.6
Claude-sonnet-4.5	58.4	53.6	52.4
Gemini-2.5-pro	63.4	<u>68.1</u>	<u>62.2</u>
InternVL-3.5-8B	28.8	9.1	10.9
MiMo-VL-7B	21.6	24.5	20.0
GLM-4.6V-Flash	31.0	23.4	21.3
Qwen3-VL-8B-Instruct	28.0	18.0	17.7
Ours(RailMind-8B)	62.0	59.7	56.8

GLM-4.5 attain 46.7% and 45.6%, respectively—indicating strong *semantic understanding and risk-category analysis* of railway anomalies when the task is cast as category recognition rather than localization. This stands in sharp contrast to Table I, where the same APIs achieve near-vanishing detection PR/RC; the gap confirms that high F1 on classification does not translate into reliable spatial grounding under IoU. Among comparable-scale open-source models ($\sim 7\text{B}$ – 9B), generic instruction-tuned baselines cluster at roughly 10–31% $F1_M$, whereas **RailMind-8B** reaches **56.8%**. This result shows that RailMind outperforms same-scale alternatives and substantially narrows the margin to much larger models, which we attribute to domain-aligned instruction tuning that better captures railway-specific defect semantics and security implications under long-tailed, safety-critical conditions.

C. Ablation Study

TABLE III
ABLATION ON RL POST-TRAINING: MACRO PR/RC/F1 FOR IMAGE-LEVEL SECURITY-RELEVANT ANOMALY CLASSIFICATION.

Models	PR_M	RC_M	$F1_M$
RailMind-8B-SFT	48.7	62.5	52.5
RailMind-8B-SFT+GRPO	59.8	58.2	54.8
RailMind-8B-SFT+GSPO	62.0	59.7	56.8

Table III ablates RL post-training on top of RailMind-8B-SFT using the same image-level category-set evaluation as Table II (Sec. IV-B), *i.e.*, only *which* security-relevant defect categories are predicted per image, without box matching. Starting from SFT at $PR_M=48.7\%$, $RC_M=62.5\%$, and $F1_M=52.5\%$, adding GRPO lifts precision to 59.8% but lowers recall to 58.2%, yielding $F1_M=54.8\%$. GSPO reaches the strongest PR_M and $F1_M$ overall (62.0% / 59.7% / 56.8%) by improving precision over SFT while recovering recall compared with GRPO (59.7% vs. 58.2%). In short, reward-driven post-training trades off precision and recall in different ways across GRPO and GSPO, with GSPO offering the best macro F1 for image-level security screening in our setting.

Table IV isolates the primary MLLM while fixing the Grounding-DINO detector and its training recipe: only the text prompts from RailMind-8B-SFT, SFT+GRPO, or SFT+GSPO

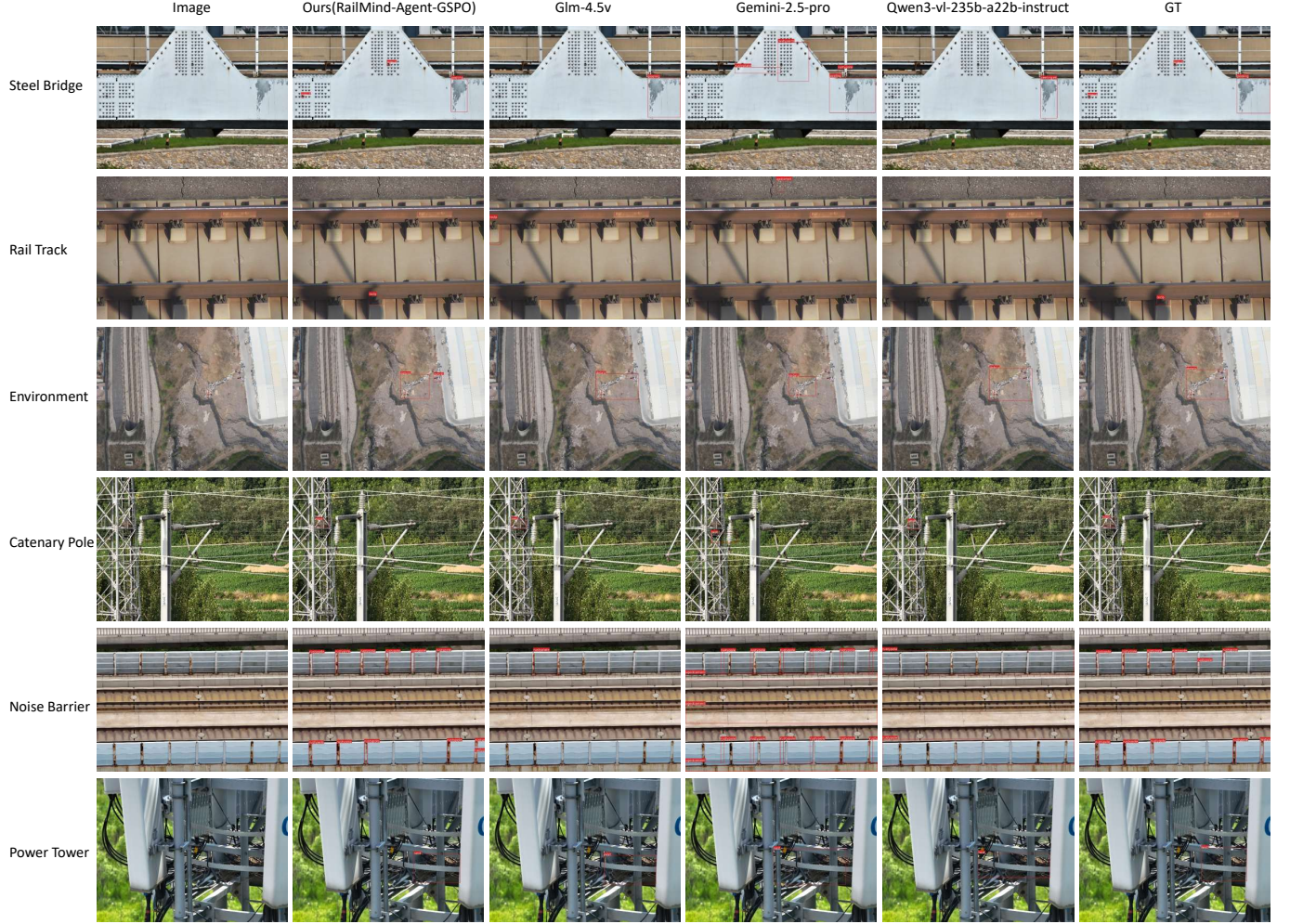


Fig. 6. Qualitative comparison of different models. Our RailMind agent demonstrates superior security-relevant anomaly localization across the six major railway scenarios and 13 defect categories.

TABLE IV
MACRO PR/RC/F1 OF GROUNDING-DINO ANOMALY LOCALIZATION
UNDER DIFFERENT PRIMARY MLLM AGENTS (FIXED DETECTOR).

Models	PR_M	RC_M	$F1_M$
RailMind-8B-SFT	61.9	42.4	50.3
RailMind-8B-SFT+GRPO	65.7	43.5	52.3
RailMind-8B-SFT+GSPO	66.6	42.2	51.7

change. Relative to SFT ($PR_M=61.9\%$, $RC_M=42.4\%$, $F1_M=50.3\%$), both RL stages raise PR_M (to 65.7% / 66.6%), indicating that tighter language–category alignment reduces spurious detections and improves prompt-conditioned grounding for security evidence localization. RC_M remains in a narrow band (42.2% – 43.5%), which suggests that recall is still dominated by localization and sparse labels rather than by prompt wording alone. GRPO achieves the highest recall and $F1_M$ (43.5% / 52.3%), whereas GSPO peaks in PR_M (66.6%) with slightly lower recall (42.2%) and $F1_M$ (51.7%); we

adopt SFT+GSPO as the default primary agent when reporting detection in the main tables, following the same choice as for image-level classification.

TABLE V
MACRO PR/RC/F1 OF RAILMIND-8B-SFT+GSPO + GROUNDING-DINO
UNDER DIFFERENT IOU THRESHOLDS FOR ANOMALY EVIDENCE
LOCALIZATION.

IoU	PR_M	RC_M	$F1_M$
0.1	73.5	45.8	56.4
0.2	71.9	45.0	55.4
0.3	70.0	44.0	54.0
0.4	68.6	43.3	53.1
0.5	66.6	42.2	51.7

Table V uses the same RailMind-8B-SFT+GSPO primary agent and Grounding-DINO setup as Table IV, but varies the IoU cutoff τ for box–box matching (Sec. IV-B). Raising τ from 0.1 to 0.5 monotonically lowers PR_M , RC_M , and $F1_M$ (from 73.5% / 45.8% / 56.4% down to 66.6% / 42.2% / 51.7%),

because stricter overlap penalizes small and irregular defects in UAV views. This trend shows that RailMind remains robust as the matching criterion becomes stricter; we use $\tau=0.5$ for the main comparison in Table I to report stricter localization performance suitable for inspection evidence review.

D. Qualitative Results

Figure 6 provides qualitative comparisons under the strict IoU = 0.5 setting. Overall, RailMind produces more accurate and complete localizations than strong multimodal baselines, especially for small, sparse, or heavily cluttered defects that require reliable evidence for security inspection. These visual results are consistent with the quantitative gains reported in Table I.

In the Steel Bridge and Rail Track scenes, RailMind successfully localizes challenging defects such as *missing nut* and *missing fastener*, whereas the baseline MLLMs either miss them or return loose boxes. By contrast, the relatively more salient *coating peeling* defect can be detected by all models, suggesting that the main difficulty lies in subtle, small-scale anomalies requiring stronger domain knowledge and fine-grained grounding. A similar trend appears in the Environment scene: although all methods can identify *trash*, RailMind yields denser and tighter predictions, leading to visibly better recall on scattered instances that may affect surrounding railway safety.

The advantages of RailMind become more evident in the Catenary Pole, Power Tower, and Noise Barrier scenes. For *nest* detection, the compared MLLMs can often recognize the anomaly category but still exhibit inaccurate spatial grounding, such as severe offsets or incomplete coverage; in contrast, RailMind generates boxes that more closely match the true defect extent. The Noise Barrier scene is particularly challenging because *pillar rust* appears as many small, dispersed regions within complex backgrounds. Here, RailMind is able to identify nearly all affected pillars with precise localization, while the baselines suffer from missed detections, coarse boxes, or substantial offsets. These cases further support our claim that decoupling semantic reasoning from a dedicated grounding module is crucial for reliable railway anomaly localization and trustworthy security evidence extraction.

VI. CONCLUSION

In this work, we presented RailMind, a lightweight railway UAV inspection framework that unifies domain-aware multimodal reasoning with fine-grained visual grounding. By coupling a distilled MLLM agent with a Grounding-DINO detector, and further supporting training with structured data generation, Banana-based augmentation, and RL post-training, RailMind provides a practical solution for open-vocabulary anomaly understanding and localization in complex railway scenes. We also established a multi-scene railway UAV benchmark with aligned protocols for both detection and image-level classification. Extensive experiments show that RailMind achieves state-of-the-art detection performance on this benchmark and strong classification capability with only 8B parameters, while remaining competitive with powerful closed-source

models in semantic understanding. We hope this benchmark and lightweight pipeline will support future research on safe, scalable, and interpretable railway inspection.

REFERENCES

- [1] G. M. Makrakis, C. Kolias, G. Kambourakis, C. Rieger, and J. Benjamin, "Industrial and critical infrastructure security: Technical analysis of real-life security incidents," *IEEE Access*, vol. 9, pp. 165 295–165 325, 2021.
- [2] J. Hua, Z. Shen, and S. Zhong, "We can track you if you take the metro: Tracking metro riders using accelerometers on smartphones," *IEEE Transactions on Information Forensics and Security*, vol. 12, no. 2, pp. 286–297, 2017.
- [3] C. Xu, J. Shen, and X. Du, "A method of few-shot network intrusion detection based on meta-learning framework," *IEEE Transactions on Information Forensics and Security*, vol. 15, pp. 3540–3552, 2020.
- [4] J. Meng, Z. Zhang, H. Zhu, Z. Yang, R. Deng, and P. Cheng, "Sstaf: Security settings-based threat assessment framework of programmable logic controllers," *IEEE Transactions on Information Forensics and Security*, vol. 20, pp. 7512–7527, 2025.
- [5] S. Soderi, D. Masti, and Y. Z. Lun, "Railway cyber-security in the era of interconnected systems: A survey," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 7, pp. 6764–6779, 2023.
- [6] S. Kim, Y. Won, I.-H. Park, Y. Eun, and K.-J. Park, "Cyber-physical vulnerability analysis of communication-based train control," *IEEE Internet of Things Journal*, vol. 6, no. 4, pp. 6353–6362, 2019.
- [7] J. T. Zhou, J. Du, H. Zhu, X. Peng, Y. Liu, and R. S. M. Goh, "AnomalyNet: An anomaly detection network for video surveillance," *IEEE Transactions on Information Forensics and Security*, vol. 14, no. 10, pp. 2537–2550, 2019.
- [8] H. Yuan, Z. Cai, H. Zhou, Y. Wang, and X. Chen, "Transanomaly: Video anomaly detection using video vision transformer," *IEEE Access*, vol. 9, pp. 123 977–123 986, 2021.
- [9] R. S. Pahwa, J. Chao, J. Paul, Y. Li, M. T. L. Nwe, S. Xie, A. James, A. Ambikapathi, Z. Zeng, and V. R. Chandrasekhar, "Faultnet: Faulty rail-valves detection using deep learning and computer vision," in *2019 IEEE Intelligent Transportation Systems Conference (ITSC)*. IEEE, 2019, pp. 559–566.
- [10] M. Wang, K. Li, X. Zhu, and Y. Zhao, "Detection of surface defects on railway tracks based on deep learning," *IEEE Access*, vol. 10, pp. 126 451–126 465, 2022.
- [11] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "Yolov4: Optimal speed and accuracy of object detection," *arXiv preprint arXiv:2004.10934*, 2020.
- [12] Z. Cao, Y. Qin, Z. Xie, Q. Liu, E. Zhang, Z. Wu, and Z. Yu, "An effective railway intrusion detection method using dynamic intrusion region and lightweight neural network," *Measurement*, vol. 191, p. 110564, 2022.
- [13] L. Guan, L. Jia, Z. Xie, and C. Yin, "A lightweight framework for obstacle detection in the railway image based on fast region proposal and improved yolo-tiny network," *IEEE Transactions on Instrumentation and Measurement*, vol. 71, pp. 1–16, 2022.
- [14] O. Zendel, M. Murschitz, M. Zeilinger, D. Steininger, S. Abbasi, and C. Beleznaï, "Railsem19: A dataset for semantic rail scene understanding," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2019, pp. 0–0.
- [15] Y. Chen, N. Zhu, Q. Wu, C. Wu, W. Niu, and Y. Wang, "Mrsi: A multi-modal proximity remote sensing data set for environment perception in rail transit," *International Journal of Intelligent Systems*, vol. 37, no. 9, pp. 5530–5556, 2022.
- [16] T. Toprak, B. Belenlioglu, B. Aydin, C. Guzelis, and M. A. Selver, "Conditional weighted ensemble of transferred models for camera based onboard pedestrian detection in railway driver support systems," *IEEE Transactions on Vehicular Technology*, vol. 69, no. 5, pp. 5041–5054, 2020.
- [17] Z. Zhang, S. Yu, S. Yang, Y. Zhou, and B. Zhao, "Rail-5k: A real-world dataset for rail surface defects detection," *arXiv preprint arXiv:2106.14366*, 2021.
- [18] T. Ye, X. Cong, Y. Li, Z. Zheng, X. Deng, X. Yan, X. Li, and X. Zhang, "Few-shot railway intrusion detection without forgetting via double rpn and detector," *IEEE Transactions on Intelligent Vehicles*, 2024.
- [19] P. Zajec, J. M. Rožanec, S. Theodoropoulos, M. Fontul, E. Koehorst, B. Fortuna, and D. Mladenčić, "Few-shot learning for defect detection in manufacturing," *International Journal of Production Research*, vol. 62, no. 19, pp. 6979–6998, 2024.

- [20] R. Ferdousi, C. Yang, M. A. Hossain, F. Laamarti, M. S. Hossain, and A. E. Saddik, "Generative model-driven synthetic training image generation: An approach to cognition in railway defect detection," *Cognitive Computation*, vol. 16, no. 5, pp. 1–16, 2024.
- [21] Z. Zhang, P. Chen, Y. Huang, L. Dai, F. Xu, and H. Hu, "Railway obstacle intrusion warning mechanism integrating yolo-based detection and risk assessment," *Journal of Industrial Information Integration*, vol. 38, p. 100571, 2024.
- [22] R. D. D. Sanchez, X. D. Ortiz, X. Zhou, M. P. Ronecker, M. Karner, D. Watenig, and A. Knoll, "Lidar-guided monocular 3d object detection for long-range railway monitoring," in *2025 IEEE Intelligent Vehicles Symposium (IV)*. IEEE, 2025, pp. 261–268.
- [23] H. Liu, C. Li, Q. Wu, and Y. J. Lee, "Visual instruction tuning," *Advances in neural information processing systems*, vol. 36, pp. 34 892–34 916, 2023.
- [24] H. Liu, C. Li, Y. Li, and Y. J. Lee, "Improved baselines with visual instruction tuning," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2024, pp. 26 296–26 306.
- [25] J. Bai, S. Bai, Y. Chu, Z. Cui, K. Dang, X. Deng, Y. Fan, W. Ge, Y. Han, F. Huang *et al.*, "Qwen technical report," *arXiv preprint arXiv:2309.16609*, 2023.
- [26] P. Wang, S. Bai, S. Tan, S. Wang, Z. Fan, J. Bai, K. Chen, X. Liu, J. Wang, W. Ge *et al.*, "Qwen2-vl: Enhancing vision-language model's perception of the world at any resolution," *arXiv preprint arXiv:2409.12191*, 2024.
- [27] OpenAI, "Hello gpt-4o," <https://openai.com/index/hello-gpt-4o/>, 2024.
- [28] S. Bai, Y. Cai, R. Chen, K. Chen, X. Chen, Z. Cheng, L. Deng, W. Ding, C. Gao, C. Ge, W. Ge, Z. Guo, Q. Huang, J. Huang, F. Huang, B. Hui, S. Jiang, Z. Li, M. Li, M. Li, K. Li, Z. Lin, J. Lin, X. Liu, J. Liu, C. Liu, Y. Liu, D. Liu, S. Liu, D. Lu, R. Luo, C. Lv, R. Men, L. Meng, X. Ren, X. Ren, S. Song, Y. Sun, J. Tang, J. Tu, J. Wan, P. Wang, P. Wang, Q. Wang, Y. Wang, T. Xie, Y. Xu, H. Xu, J. Xu, Z. Yang, M. Yang, J. Yang, A. Yang, B. Yu, F. Zhang, H. Zhang, X. Zhang, B. Zheng, H. Zhong, J. Zhou, F. Zhou, J. Zhou, Y. Zhu, and K. Zhu, "Qwen3-vl technical report," *arXiv preprint arXiv:2511.21631*, 2025.
- [29] Anthropic, "Claudeai," 2025. [Online]. Available: <https://claude.ai/>
- [30] D. Guo, F. Wu, F. Zhu, F. Leng, G. Shi, H. Chen, H. Fan, J. Wang, J. Jiang, J. Wang *et al.*, "Seed1. 5-vl technical report," *arXiv preprint arXiv:2505.07062*, 2025.
- [31] G. Comanici, E. Bieber, M. Schaeckermann, I. Pasupat, N. Sachdeva, I. Dhillon, M. Blistein, O. Ram, D. Zhang, E. Rosen *et al.*, "Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities," *arXiv preprint arXiv:2507.06261*, 2025.
- [32] X. Wei, "Duomo/transgpt," 2023. [Online]. Available: <https://github.com/DUOMO/TransGPT>
- [33] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark *et al.*, "Learning transferable visual models from natural language supervision," in *International conference on machine learning*. PmlR, 2021, pp. 8748–8763.
- [34] A. Zareian, K. D. Rosa, D. H. Hu, and S.-F. Chang, "Open-vocabulary object detection using captions," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 14 393–14 402.
- [35] Y. Zhong, J. Yang, P. Zhang, C. Li, N. Codella, L. H. Li, L. Zhou, X. Dai, L. Yuan, Y. Li *et al.*, "Regionclip: Region-based language-image pretraining," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 16 793–16 803.
- [36] L. H. Li, P. Zhang, H. Zhang, J. Yang, C. Li, Y. Zhong, L. Wang, L. Yuan, L. Zhang, J.-N. Hwang *et al.*, "Grounded language-image pre-training," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 10 965–10 975.
- [37] Z. Wang, Y. Li, X. Chen, S.-N. Lim, A. Torralba, H. Zhao, and S. Wang, "Detecting everything in the open world: Towards universal object detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 11 433–11 443.
- [38] X. Zhou, R. Girdhar, A. Joulin, P. Krähenbühl, and I. Misra, "Detecting twenty-thousand classes using image-level supervision," in *European conference on computer vision*. Springer, 2022, pp. 350–368.
- [39] S. Liu, Z. Zeng, T. Ren, F. Li, H. Zhang, J. Yang, Q. Jiang, C. Li, J. Yang, H. Su *et al.*, "Grounding dino: Marrying dino with grounded pre-training for open-set object detection," in *European conference on computer vision*. Springer, 2024, pp. 38–55.
- [40] F. Li, H. Zhang, S. Liu, J. Guo, L. M. Ni, and L. Zhang, "Dn-detr: Accelerate detr training by introducing query denoising," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 13 619–13 627.
- [41] S. Liu, F. Li, H. Zhang, X. Yang, X. Qi, H. Su, J. Zhu, and L. Zhang, "Dab-detr: Dynamic anchor boxes are better queries for detr," *arXiv preprint arXiv:2201.12329*, 2022.
- [42] H. Zhang, F. Li, S. Liu, L. Zhang, H. Su, J. Zhu, L. M. Ni, and H.-Y. Shum, "Dino: Detr with improved denoising anchor boxes for end-to-end object detection," *arXiv preprint arXiv:2203.03605*, 2022.
- [43] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable detr: Deformable transformers for end-to-end object detection," *arXiv preprint arXiv:2010.04159*, 2020.
- [44] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proceedings of the IEEE international conference on computer vision*, 2017, pp. 2980–2988.
- [45] H. Rezatofighi, N. Tsoi, J. Gwak, A. Sadeghian, I. Reid, and S. Savarese, "Generalized intersection over union: A metric and a loss for bounding box regression," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 658–666.
- [46] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 10 684–10 695.
- [47] T. Brooks, A. Holynski, and A. A. Efros, "Instructpix2pix: Learning to follow image editing instructions," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2023, pp. 18 392–18 402.
- [48] B. Yang, S. Gu, B. Zhang, T. Zhang, X. Chen, X. Sun, D. Chen, and F. Wen, "Paint by example: Exemplar-based image editing with diffusion models," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2023, pp. 18 381–18 391.
- [49] X. Chen, L. Huang, Y. Liu, Y. Shen, D. Zhao, and H. Zhao, "Anydoor: Zero-shot object-level image customization," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2024, pp. 6593–6602.
- [50] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, and S. Belongie, "Class-balanced loss based on effective number of samples," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 9268–9277.
- [51] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.
- [52] Z. Shao, P. Wang, Q. Zhu, R. Xu, J. Song, X. Bi, H. Zhang, M. Zhang, Y. Li, Y. Wu *et al.*, "Deepseekmath: Pushing the limits of mathematical reasoning in open language models," *arXiv preprint arXiv:2402.03300*, 2024.
- [53] C. Zheng, S. Liu, M. Li, X.-H. Chen, B. Yu, C. Gao, K. Dang, Y. Liu, R. Men, A. Yang *et al.*, "Group sequence policy optimization," *arXiv preprint arXiv:2507.18071*, 2025.